

Edouard Murat

Gameplay Programmer & Team Lead

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French: fluent • English: fluent

PROFILE

Gameplay programmer and team lead with 20 years of software experience, including five years of production C#. Led a six-person Godot team building curriculum-aligned educational games for Android. Specializes in gameplay systems, AI behavior, procedural generation, shader programming, and reusable game frameworks. Shipped games on Steam, Google Play, and the Apple Store.

TECHNICAL SKILLS

Game Engines	Godot, Unity
Languages	GScript, C#, Python, Java, HTML/CSS/JS, SQL
Shaders & Graphics	Shader scripting (Godot, Unity)
Gameplay Systems	Input (keyboard & mobile), AI behaviors, procedural generation, physics-based mechanics, state machines, combat systems
Tools & Frameworks	Git, GitHub, Blender, VS Code, Django, SQLite, Audacity
Other	Agile workflows, UI integration, performance optimization

EXPERIENCE

Lead Gameplay Programmer (Godot) — Sigma Digital, South Africa (remote) Sept 2025 – Aug 2026

Led a six-person Godot team building curriculum-aligned literacy games delivered on Android tablets to first- and second-grade children in Africa.

- Hired the programming team and led it as technical lead, owning architecture, code review, and sprint planning
- Designed the platform's UI framework and an MVVM + FSM architecture adopted across all activities — decoupling game logic from presentation, enabling per-activity unit testing, and making curriculum content swappable without rebuilding core mechanics
- Delivered six mini-games and supported two rounds of playtesting with the target age group
- Built for low-cost Android tablets, with gameplay mapped to a first- and second-grade reading and writing curriculum
- Worked directly with educators and instructional designers to translate curriculum goals into game mechanics, bridging educational theory and interactive design across each sprint

Game Developer — EdNoKa, Montreal 2022 – Present

Founder and lead developer of an educational gaming platform blending custom-built quizzes with gameplay across farming, hack & slash, platformer, puzzle, and sports genres.

- Published 5+ games as public Steam playtests using Unity and Godot, integrating real-time quizzes into diverse gameplay loops
- Designed and implemented gameplay systems: player control, AI, procedural generation, level logic, scoring, and feedback
- Implemented unified input systems supporting both keyboard and mobile touch controls
- Created custom shaders for visual storytelling and UX (X-ray vision, mini-cameras, 3D text, underwater effects)
- Built a full-stack Django backend: authentication, data storage, in-game unlockables, and content delivery
- Built and maintained deployment pipelines for Steam, Google Play, and the Apple Store; games live on the public Steam playtest

Web Developer and Graphic Designer — HeadPause, Montreal 2018 – 2020

- Developed and maintained the company website and online presence
- Created animations and visual assets, and ran social advertising campaigns

Java and Web Developer — BMC Software, Montreal 2013 – 2014

- Partial management of the development team in SCRUM, including code reviews and TDD
- Built a new adaptive website to client requirements alongside parallel remote teams
- Interviewed candidates for web developer positions

C# Developer and Analyst — Fujitsu Consulting, Montreal 2008 – 2013

- Five years of production C#: a payment solution for HydroSherbrooke and the daycare management system for the Ministère de la Famille du Québec
- Analyzed client needs, resolved blocking issues, and trained new developers across a three-year project

SELECTED GAME PROJECTS

Full list of playable games at edouardmurat.com

Base of 30 (Godot) — Rogue hack & slash through procedurally generated rooms 2025

- Designed and implemented all core gameplay systems, including combat and player progression
- Developed procedural generation for room layouts, obstacle placement, and navigation paths
- Built custom behavior trees for enemy types with unique attack patterns and movement logic

Aerial Ambush (Godot) — Skill-based platformer with homing missiles

2025

- Implemented responsive, high-speed player movement with precise collision handling
- Developed homing missile AI with adjustable targeting and physics-based steering
- Designed and balanced dodge-based gameplay loops for short sessions and high replayability

Agent Ka (Godot) — Waldo-like game in a generated city with crowds and powers

2024

- Built a procedural city generation system with randomized buildings, roads, and crowds
- Programmed crowd simulation with dynamic movement, behaviors, and density-based logic
- Implemented X-ray, mini-camera, and slow-motion powers via custom shaders

Veggies Island (Godot) — Farming game on a cartoonish island

2024

- Implemented the core farming loop: planting, watering, harvesting, and crop growth stages
- Built procedural island generation with randomized terrain, resources, and farm zones
- Developed a lightweight inventory system for harvested crops and seeds

Pico Golf (Godot) — Mini golf challenges with physics-based terrain

2023

- Designed 3D mini-golf courses with increasing difficulty and varied sea-floor layouts
- Integrated animated fish and custom water and caustics shaders to enhance immersion
- Developed audio-visual feedback systems for a calming, low-pressure experience

Tomb (Godot) — Doom-inspired shooter for Godot Wild Jam #42

2022

- Built from scratch in under one week during a jam, demonstrating rapid prototyping
- Engineered arcade-style shooter feel: player movement, shooting, and health systems
- Designed enemy AI using state machines and pathfinding logic with trap systems

Earlier titles include Left 4 Dead 2D, Meditation Card Game, Fruit, Archer, and Meditation World VR.

EDUCATION

B.A. Psychology — Concordia University, Montreal	2015 – 2017
B.Eng. Computer Engineering — University of Sherbrooke	2005 – 2009
College Diploma, DEC in Sciences — Stanislas College, Montreal	2003 – 2005

CERTIFICATIONS

Certified Associate Python Programmer — Python Institute	2024
Oracle Professional Certification, Java SE 7 Programmer	2013
Oracle Associate Certification, Java SE 7 Programmer	2012

References available upon request.